



1.4.5 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.



Unit 1, Lesson 5: Solving Mathematical and Real-World Problems With Equations



Benchmark

MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical context or real-world context, where all terms are rational numbers.



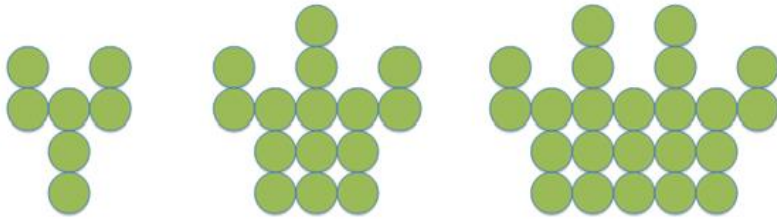
Learning Targets

- ☐ I can solve two-step equations involving the distributive property.
- ☐ I can solve two-step equations within a real-world context.
- ☐ I can relate the solution of an equation representing a real-world context to the context.



1.5.1 Warm-Up: Number Patterns

A pattern is shown.



1. Describe how the pattern is changing in each step.
2. How many circles will be in the next step?



1.5.2 Refresher: The Distributive Property

1. Several expressions are shown.

$$(-2 \cdot x) + 3$$

$$(x + 3)(-2)$$

$$-2x - 6$$

$$(-2) \cdot x + (-2) \cdot 3$$

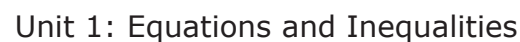
$$(-2 + x)3$$

- a. Circle the expressions that are equivalent to $-2(x + 3)$.
 - b. Use substitution to show that each of the expressions chosen is equivalent to $-2(x + 3)$.
2. Rewrite each expression by applying the distributive property.
 - a. $(-x + 6)4$
 - b. $-3.5(-7 + 2y)$
 - c. $\frac{7}{8}\left(16m - \frac{4}{5}\right)$



1. The equation $65 = 8(9 - p)$ can be solved using either method.
 - a. Solve the equation $65 = 8(9 - p)$ using each method.

- b. Verify the solution is correct using substitution.
- c. Discuss each method of solving the equation with your partner. Then, explain which method you preferred for solving this equation.

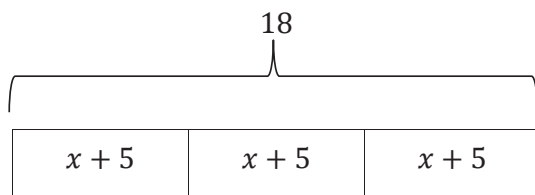




1.5.4 Collaboration: Matching Equations to Diagrams and Contexts

Equations can be represented using bar diagrams.

1. A bar diagram is shown.



- a. Select all the equations that match the diagram.

- ☐ $x + 5 = 18$
- ☐ $18 \div 3 = x + 5$
- ☐ $3(x + 5) = 18$
- ☐ $x + 5 = \frac{1}{3} \cdot 18$
- ☐ $3x + 5 = 18$

- b. Compare your selections with your partner. Resolve any differences if necessary.

- c. With your partner, choose one of the equations that can be modeled by the bar diagram and explain why it matches the diagram.

- d. Explain why $x + 5 = 18$ does not match the bar diagram.

Equations can be used to represent real-world contexts.

2. Three equations are shown.

Equation A	Equation B	Equation C
$\frac{1}{4} + 4x = 8$	$4 + \frac{1}{4}x = 8$	$8x + \frac{1}{4} = 4$

- a. Work with your partner to determine which equation matches each context. Then, write each equation in the appropriate row.

Context	Equation
A stack of nested paper cups is 8 inches tall. The first cup is 4 inches tall and each of the rest of the cups in the stack adds $\frac{1}{4}$ inch to the height of the stack.	
A baker uses 4 cups of flour. She uses $\frac{1}{4}$ cup to cover the counter with flour so the dough doesn't stick, and the rest to make 8 identical loaves of bread.	
Elena has an 8-foot piece of ribbon. She cuts off a piece that is $\frac{1}{4}$ of a foot long and cuts the remainder into four pieces of equal length.	

- b. With your partner, discuss what the variable x represents in each of the equations. Then, write the meaning of x in each equation based on the context it matches.

Equation	Meaning of x
A: $\frac{1}{4} + 4x = 8$	
B: $4 + \frac{1}{4}x = 8$	
C: $8x + \frac{1}{4} = 4$	



1.5.5 Guided Instruction: Relating Equations to Their Contexts

- Franco is saving money to take a trip to Cocoa Beach, Florida with his friends. Every week, he sets aside the same portion of his paycheck after taking out the \$325 he needs for his weekly expenses. After 11 weeks, Franco has \$489.94 saved for his trip.
 - An equation can be created relating these quantities as described below.

$$\left(\begin{array}{c} \text{Number} \\ \text{of} \\ \text{weeks} \end{array} \right) \left(\begin{array}{c} \text{Amount} \\ \text{of} \\ \text{weekly} \\ \text{paycheck} \end{array} - \begin{array}{c} \text{Amount} \\ \text{of weekly} \\ \text{expenses} \end{array} \right) = \begin{array}{c} \text{Total} \\ \text{Saved} \end{array}$$

Which of these quantities is unknown?

- Write the equation that represents this situation using the variable p for the unknown quantity.
- What does $(p - 325)$ represent in the equation?

d. Solve the equation.

e. Explain what the solution means in this context.



1.5.6 Try It: Relating Equations to Their Contexts

Work with your partner to complete the following.

1. Two equations are shown.

a. Solve both equations. Show your work.

Equation A	Equation B
$46.92 = 6(y - 1.75)$	$6y - 1.75 = 46.92$

b. A family of 6 is going to the local fair. They have a coupon for \$1.75 off of each ticket. They pay \$46.92 for all their tickets. Explain which equation represents this situation.

c. Explain what the solution to the equation that correctly models this situation means.



1.5.7 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.

Unit 1, Lesson 6: Writing Two-Step Equations From Real-World Contexts



Benchmark

MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical context or real-world context, where all terms are rational numbers.



Learning Targets

- ☐ I can write two-step equations to represent real-world contexts.
- ☐ I can solve two-step equations within a real-world context.